

Task 1: Real-World Example of Cyber Threats

Threat Type	Example	How the Attack Works
Phishing	Fake Paytm message	An attacker sends a fake email/SMS pretending to be Paytm and asks the user to verify an account through a fraudulent website.
Phishing	Fake WhatsApp login	A victim receives a fake WhatsApp verification message and is directed to a fraudulent page designed to steal login or verification information.
Phishing	Fake Flipkart offer	An attacker sends a fake discount or prize message containing a fraudulent link that attempts to trick the user into providing personal information.
Ransomware	WannaCry	Malware encrypts files on an infected computer and demands payment from the victim to restore access.
Ransomware	LockBit	Attackers use ransomware to encrypt organizational data and may also threaten to publish stolen information.
Ransomware	Ryuk	Ryuk has been used against organizations to encrypt important files and disrupt business operations until a ransom is demanded.
Insider Threat	Employee copying	An employee with legitimate

	confidential data	access may copy sensitive company information to a personal device or unauthorized storage.
Insider Threat	Misuse of database access	An authorized employee may access customer records beyond what is required for their job.
Insider Threat	Disgruntled employee	A dissatisfied employee may intentionally delete, modify, or leak organizational information using their legitimate access.

Task 2: Simulated Phishing Page in a Controlled Environment

Steps

- setoolkit
- Social-Engineering Attacks
 - Website Attart Vectors
 - Credential Harvester Attack Method
 - Site Cloner

Task 3: Email Spoofing Demonstration

Setup:

- `sudo apt install sendmail sendmail-bin`
- `sudo sendmailconfig`
- `sudo service sendmail start`

Send a spoofed test to your own address (local MTA):

Method 1: raw sendmail

`sendmail -t -i <<'EOF'`

To: your-own-address@gmail.com

From: "BookMyShow Support" <support@bookmyshow.com>

Subject: Your ticket is confirmed - Order #BK123456

Dear customer, this is a spoofing test in a controlled lab. Please ignore.
EOF

```
# Method 2: set envelope sender explicitly with -f  
sendmail -f "IRCTC Alerts <no-reply@irctc.co.in>"  
your-own-address@gmail.com <<'EOF'  
Subject: E-Ticket PNR 1234567890
```

This message was sent from your local sendmail to demonstrate spoofing.
EOF

Task 4: Prevention Techniques Against Phishing and Email Spoofing

Technique 1: Enable Two-Factor Authentication

- Two-factor authentication (2FA) adds another security layer in addition to a password.

Even if an attacker obtains a password through phishing, the attacker may still be unable to access the account without the second authentication factor.

Technique 2: Check Email Headers and Links

Task 5: How a Phishing Attack Could Target a Zomato or Swiggy User

Example Attack Scenario

An attacker could send a fake message claiming:

"Your food delivery account has been suspended. Verify your account immediately to continue using the service."

The message could contain a link leading to a fraudulent website that visually resembles a food-delivery application.

The attacker may try to convince the victim to enter information such as a username, password, OTP, or payment information.

Step-by-Step Guide to Recognize and Avoid the Attack

Step 1 – Check the sender

Look carefully at the email address, phone number, or account that sent the message.

Step 2 – Check the link

Do not immediately click a link. Check whether the destination belongs to the legitimate service.

Step 3 – Look for urgency

Messages saying "your account will be deleted in 10 minutes" or "verify within 24 hours" are common social-engineering techniques.

Step 4 – Never share OTPs

A legitimate support representative should not ask you to disclose an OTP received on your phone.

Step 5 – Open the official application

Instead of using a link received in an unexpected message, open the official food-delivery application directly.

Step 6 – Report suspicious messages

Use the email provider or application reporting mechanism to report suspected phishing.

Three Major Warning Signs

1. Suspicious or misspelled website domain
2. Urgent threats or unrealistic offers
3. Requests for passwords, OTPs, card information, or other sensitive information